INJECTIVITY OF THE SOMEKAWA GALOIS SYMBOL FOR A JACOBIAN AND ANY NUMBER OF MULTIPLICATIVE GROUPS

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Allowed standard prerequisites (editorial).

- Automorphic tensor products, induction/base change and analytic Rankin-Selberg theory. Original complete bibliography and source-local auxiliary L-function arguments retained.

1. Introduction

Let $G_1, \dots, G_s$ be semi-abelian varieties over a field k. Somekawa introduced the K-group $K(k; G_1, \dots, G_s)$, now called the Somekawa K-group, by generators over finite extensions of k and by projection-formula and reciprocity relations [Som90]. If n is prime to the characteristic $\mathrm{char}(k)$ of k, he also defined the Galois symbol map

$$S_n(k; G_1, \dots, G_s) \colon K(k; G_1, \dots, G_s)/n \to H^s_{\text{ét}}\big(k, G_1[n] \otimes \cdots \otimes G_s[n]\big);$$

see [Som90, Lemma 1.6]. Somekawa conjectured that this map is injective [Som90, Remark 1.7]. The conjecture is not true for all semi-abelian varieties; see [SY09]. It is therefore useful to find natural classes for which the symbol is injective.

Let C be a smooth projective geometrically connected curve over k with a k-rational point, and let $J = \mathrm{Jac}_C$ be its Jacobian. Spiess proved that

$$S_n(k; \mathbb{G}_m, J) \colon K(k; \mathbb{G}_m, J)/n \to H^2_{\text{ét}}(k, \mu_n \otimes J[n])$$

is injective ([Yam05, Appendix, Theorem 6.1]). His proof uses the Bloch–Ogus theory and the Merkurjev–Suslin theorem. The purpose of this note is to extend this result to an arbitrary number of copies of $\mathbb{G}_m$.

Theorem 1.1. *Let k be a field, and let C be a projective smooth geometrically connected curve over k with a k-rational point. Let J be the Jacobian variety of C. Let $r \geq 1$, and let n be a positive integer prime to the characteristic* $\mathrm{char}(k)$ *of k. Then the Galois symbol map*

$$S_n^{J,r} \colon K(k; J, \underbrace{\mathbb{G}_m, \dots, \mathbb{G}_m}_{r})/n \to H^{r+1}_{\text{ét}}\big(k, J[n] \otimes \mu_n^{\otimes r}\big)$$

is injective.

The permutation of factors sends $\{z, a_1, \dots, a_r\}_{E/k}$ to $\{a_1, \dots, a_r, z\}_{E/k}$. After the corresponding permutation of the coefficient modules, the two Galois symbols differ by the sign $(-1)^r$. Hence the theorem is equivalent to the injectivity of

$$K(k; \underbrace{\mathbb{G}_m, \dots, \mathbb{G}_m}_{r}, J)/n \to H^{r+1}_{\text{ét}}\big(k, \mu_n^{\otimes r} \otimes J[n]\big).$$

For $r = 1$, after permuting the two factors, Theorem 1.1 gives another proof of Spiess's theorem [Yam05, Appendix, Theorem 6.1]. The proof reduces to two known results.

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First, Akhtar proved that a mixed K-group attached to $CH_0(C)$ and r copies of $\mathbb{G}_m$ is naturally isomorphic to the higher Chow group $CH^{r+1}(C, r)$ ([Akh04, Theorem 6.1]). A rational point gives a decomposition of this mixed K-group into

$$K_r^M(k) \oplus K(k; J, \mathbb{G}_m, \dots, \mathbb{G}_m).$$

Second, the Beilinson–Lichtenbaum comparison theorem implies that the finite coefficient étale cycle map in bidegree $(r+2, r+1)$ is injective ([GL01, Corollary 1.2], [Voe11, Theorem 6.17]).

Notation. Throughout this paper, a curve over a field means an integral separated scheme of finite type over that field of dimension one.

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2. Preliminaries

Let C be a smooth projective geometrically connected curve over k with $P \in C(k)$, and $J = \mathrm{Jac}_C$ be the Jacobian of C.

2.1. **Mixed K-groups and higher Chow groups.** For every field extension E/k, we write P_E for the base change of P. In [Akh04], Akhtar defined the mixed K-group

$$K_r\big(k; CH_0(C), \mathbb{G}_m\big) := K\big(k; CH_0(C), \underbrace{\mathbb{G}_m, \dots, \mathbb{G}_m}_{r}\big).$$

It is generated by symbols $\{ z, a_1, \dots, a_r \}_{E/k}$, where E/k is finite, $z \in CH_0(C_E)$, and $a_i \in E^\times$. The defining relations are the projection-formula and reciprocity relations; see the definition of the mixed K-group in [Akh04, Section 3]. We also define

$$K_r(k; J, \mathbb{G}_m) := K(k; J, \underbrace{\mathbb{G}_m, \dots, \mathbb{G}_m}_{r}).$$

Let n be an integer prime to $\mathrm{char}(k)$. For a finite extension E/k, let

$$\delta_E \colon E^\times/(E^\times)^n \to H^1_{\text{ét}}(E, \mu_n)$$

and

$$\delta_{J,E} \colon J(E)/nJ(E) \to H^1_{\text{ét}}(E, J[n])$$

be the Kummer connecting homomorphisms. The Galois symbol map

$$S_n^{J,r} \colon K_r(k; J, \mathbb{G}_m)/n \to H^{r+1}_{\text{ét}}\big(k, J[n] \otimes \mu_n^{\otimes r}\big)$$

is induced on symbols by

$$S_n^{J,r}\big(\{ z, a_1, \dots, a_r \}_{E/k}\big) = \mathrm{Cor}_{E/k}\big(\delta_{J,E}(z) \cup \delta_E(a_1) \cup \cdots \cup \delta_E(a_r)\big); \tag{2.1}$$

see [Som90, Lemma 1.6]. The point P gives, for every finite extension E/k, the decomposition

$$CH_0(C_E) = \mathbb{Z}[P_E] \oplus CH_0(C_E)^0 = \mathbb{Z}[P_E] \oplus J(E).$$

By [Akh00, Corollary 5.4.5] and [Som90, Proposition 1.5], it induces a natural decomposition

$$K_r\big(k; CH_0(C), \mathbb{G}_m\big) \simeq K_r^M(k) \oplus K_r(k; J, \mathbb{G}_m). \tag{2.2}$$

We use cohomological notation for higher Chow groups. By the Nesterenko–Suslin–Totaro theorem, for every field E there is a natural isomorphism $K_r^M(E) \simeq CH^r(E, r)$ ([NS90, Tot92]). We use this isomorphism to identify these two groups. Thus the notation

$\{a_1, \dots, a_r\}$ denotes both the Milnor symbol and the corresponding class in $CH^r(E, r)$. Since C has dimension one, the group of zero-cycles in degree r is $CH^{r+1}(C, r)$.

Theorem 2.1 ([Akh04, Theorem 6.1]). *For every $r \geq 0$, there is a natural isomorphism*

$$\varphi_r \colon CH^{r+1}(C, r) \xrightarrow{\sim} K_r\big(k; CH_0(C), \mathbb{G}_m\big). \tag{2.3}$$

We describe the inverse map of φ_r in (2.3). Let E/k be finite and let $p_E \colon C_E \to C$ be the projection. Via the canonical identification

$$E^\times \simeq CH^1(E, 1),$$

we regard each $a_i \in E^\times$ as its pull-back to $CH^1(C_E, 1)$ along the structure morphism of C_E. The inverse map φ_r^{-1} is characterized by

$$\varphi_r^{-1}\big(\{z, a_1, \dots, a_r\}_{E/k}\big) = (p_E)_*\big(z \cap a_1 \cap \cdots \cap a_r\big) \tag{2.4}$$

for $z \in CH_0(C_E) = CH^1(C_E)$ and $a_i \in E^\times$. Here $\cap$ denotes the intersection product on higher Chow groups

$$\cap \colon CH^p(C_E, m) \times CH^q(C_E, n) \to CH^{p+q}(C_E, m+n),$$

and

$$(p_E)_* \colon CH^{r+1}(C_E, r) \to CH^{r+1}(C, r)$$

is the proper push-forward. Thus $z \cap a_1 \cap \cdots \cap a_r \in CH^{r+1}(C_E, r)$.

Combining (2.2) and Theorem 2.1, we obtain

$$CH^{r+1}(C, r) \simeq K_r^M(k) \oplus K_r(k; J, \mathbb{G}_m). \tag{2.5}$$

Let $\iota_{J,r}$ denote the inclusion of the second factor in (2.5). By (2.4),

$$\iota_{J,r}\big(\{z, a_1, \dots, a_r\}_{E/k}\big) = (p_E)_*\big(z \cap a_1 \cap \cdots \cap a_r\big) \tag{2.6}$$

for $z \in J(E) = CH_0(C_E)^0$. We will see the decomposition (2.5) via a decomposition of the Chow motive of C (Lemma 2.2).

2.2. **Chow motives and the action of correspondences.** We briefly recall effective Chow motives. An effective Chow motive is a pair $M = (X, \pi)$ where X is a smooth projective equidimensional variety over k of $\dim X = d$ and π is a projector in the ring $CH^d(X \times X)$ of correspondences on X. We write $h(X)$ for the Chow motive $(X, 1_X)$ where 1_X is the graph of the identity id_X.

A correspondence $\pi \in CH^d(X \times X)$ induces an endomorphism π_* of $CH^i(X, j)$ as follows. For any $z \in CH^i(X, j)$,

$$\pi_*(z) := p_{2*}(\pi \cap p_1^*(z)) \in CH^i(X, j)$$

where $p_i : X \times X \to X$ is the i-th projection. Note that 1_{X*} is the identity. For an effective Chow motive $M = (X, \pi)$, we define $CH^i(M, j) := \pi_* CH^i(X, j)$. Then we have a decomposition of $CH^i(X, j)$;

$$CH^i(X, j) = CH^i(M, j) \oplus CH^i(M', j)$$

where $M' = (X, 1_X - \pi)$.

A k-rational point $P \in X(k)$ gives the following decomposition of $h(X)$;

$$h(X) = h^0(X) \oplus h^{[1,2d-1]}(X) \oplus h^{2d}(X)$$

where

$$h^0(X) = (X, [P \times X]),$$
$$h^{[1,2d-1]}(X) = (X, 1_X - [P \times X] - [X \times P])$$
$$h^{2d}(X) = (X, [X \times P]).$$

There are isomorphisms of Chow motives

$$h^0(X) \simeq (\mathrm{Spec}(k), 1_{\mathrm{Spec}(k)}),\ h^{2d}(X) \simeq \mathbb{L}^d$$

where $\mathbb{L}$ is the Lefschetz motive. If $\dim X = 1$, we write $h^1(X)$ for $h^{[1,2d-1]}(X)$, and in this case we have the following decomposition of $CH^i(X, j)$;

$$CH^i(X, j) = CH^i(h^0(X), j) \oplus CH^i(h^1(X), j) \oplus CH^i(h^2(X), j).$$

Lemma 2.2. *For any $r \geq 0$, the isomorphism (2.3) induces isomorphisms*

$$CH^{r+1}(h^0(C), r) = 0,$$
$$CH^{r+1}(h^1(C), r) \simeq K_r(k; J, \mathbb{G}_m),\ \textit{and}$$
$$CH^{r+1}(h^2(C), r) \simeq K_r^M(k).$$

Proof. Put $\alpha := [P \times C], {}^t\alpha := [C \times P]$. We first compute $\alpha_*, {}^t\alpha_*$ on $CH^1(C)$. For $z \in CH^1(C)$, we have

$$\alpha \cap (z \times C) = 0,\ {}^t\alpha \cap (z \times C) = z \times P$$

which shows

$$\alpha_*(z) = 0,\ {}^t\alpha_*(z) = \deg(z)[P] \in \mathbb{Z}[P].$$

This shows that the kernel of ${}^t\alpha_*$ on $CH^1(C)$ is equal to the kernel of the degree map

$$\deg : CH^1(C) \to \mathbb{Z}$$

and proves the assertion in the case $r = 0$.

By (2.3) and (2.4), we take an element w of $CH^{r+1}(C, r)$ of the following form

$$w := (p_E)_*\big(z \cap a_1 \cap \cdots \cap a_r\big).$$

By the base change formula for algebraic cycles with respect to proper push-forwards and flat pull-backs, it suffices to compute $\alpha_*(w)$ in case $E = k$. Then, we have

$$\begin{aligned}\alpha_*(w) &= p_{2*}\big(\alpha \cap p_1^*(z \cap a_1 \cap \cdots \cap a_r)\big)\\ &= p_{2*}\big((\alpha \cap p_1^* z) \cap p_1^*(a_1 \cap \cdots \cap a_r)\big) = 0,\end{aligned}$$

and

$$\begin{aligned}{}^t\alpha_*(w) &= p_{2*}\big({}^t\alpha \cap p_1^*(z \cap a_1 \cap \cdots \cap a_r)\big)\\ &= p_{2*}\big(({}^t\alpha \cap p_1^* z) \cap p_1^*(a_1 \cap \cdots \cap a_r)\big)\\ &= p_{2*}\big([z \times P] \cap p_1^*(a_1 \cap \cdots \cap a_r)\big)\\ &= \deg(z)\big([P] \cap a_1 \cap \cdots \cap a_r\big)\\ &= \deg(z)\{a_1, \ldots, a_r\} \in K_r^M(k).\end{aligned}$$

Here in the last equality, we identify $CH^r(P, r)$ with $K_r^M(k)$. This shows that the kernel of ${}^t\alpha_*$ on $CH^{r+1}(C, r)$ is equal to the kernel of the map

$$CH^{r+1}(C, r) \to K_r^M(k).$$

Thus, the isomorphism (2.3) induces

$$CH^{r+1}(h^0(C), r) = 0,\ CH^{r+1}(h^1(C), r) \simeq K_r(k; J, \mathbb{G}_m),\ CH^{r+1}(h^2(C), r) \simeq K_r^M(k).$$

□

By Lemma 2.2, the inclusion $\iota_{J,r}$ in (2.6) coincides with

$$\iota_{J,r} : K_r(k; J, \mathbb{G}_m) \simeq CH^{r+1}(h^1(C), r) \hookrightarrow CH^{r+1}(C, r). \tag{2.7}$$

2.3. **Étale cycle map.** For a smooth scheme X, we use the identification

$$CH^q(X, 2q-p) = H^p_{\mathrm{M}}(X, \mathbb{Z}(q));$$

see [Blo86] and [Voe02, Theorem 1]. Let

$$\rho^{i,j}_{X,n} \colon CH^i(X, j)/n \to H^{2i-j}_{\text{ét}}\big(X, \mu_n^{\otimes i}\big)$$

be the finite coefficient étale cycle map. For $i = j+1$, its injectivity follows from [GL01, Corollary 1.2] (see also [Voe11, Theorem 6.17]). Under the Nesterenko–Suslin–Totaro isomorphism $CH^r(E, r) \simeq K^M_r(E)$, the finite coefficient étale cycle map is the norm residue isomorphism

$$K^M_r(E)/n \xrightarrow{\sim} H^r(E, \mu_n^{\otimes r}), \qquad \{a_1, \dots, a_r\} \mapsto \delta_E(a_1) \cup \cdots \cup \delta_E(a_r).$$

Moreover, under the natural isomorphisms $CH^1(X) \simeq H^1_{\text{ét}}(X, \mathbb{G}_m)$ and $CH^1(X, 1) \simeq \Gamma(X, \mathbb{G}_m)$, the cycle maps $\rho^{1,0}_{X,n}, \rho^{1,1}_{X,n}$ coincide with the Kummer connecting homomorphisms

$$\begin{aligned} H^1_{\text{ét}}(X, \mathbb{G}_m)/n &\to H^2_{\text{ét}}(X, \mu_n), \\ \Gamma(X, \mathbb{G}_m)/n &\to H^1_{\text{ét}}(X, \mu_n). \end{aligned}$$

Lemma 2.3. *Let X be a smooth projective variety over k of dimension d. For any $\alpha \in CH^d(X \times X)$, put $\rho(\alpha) := \rho^{d,0}_{X\times X,n}(\alpha)$ in $H^{2d}_{\text{ét}}(X \times X, \mu_n^{\otimes d})$. Then the following diagram commutes:*

$$\begin{array}{ccc} CH^i(X, j)/n & \xrightarrow{\rho^{i,j}_{X,n}} & H^{2i-j}_{\text{ét}}(X, \mu_n^{\otimes i}) \\ \big\downarrow{\scriptstyle \alpha_*} & & \big\downarrow{\scriptstyle \rho(\alpha)_*} \\ CH^i(X, j)/n & \xrightarrow{\rho^{i,j}_{X,n}} & H^{2i-j}_{\text{ét}}(X, \mu_n^{\otimes i}) \end{array}$$

Proof. Let $p_i : X \times X \to X$ be the i-th projection. Take $z \in CH^i(X, j)$. By the compatibility of the cycle map with proper push-forward and flat pull-back, we have

$$\begin{aligned} \rho^{i,j}_{X,n}\big(\alpha_*(z)\big) &= \rho^{i,j}_{X,n}\big(p_{2*}(\alpha \cap p_1^* z)\big) \\ &= p_{2*}\Big(\rho^{d+i,j}_{X\times X,n}\big(\alpha \cap p_1^* z\big)\Big) \\ &= p_{2*}\big(\rho(\alpha) \cup \rho^{i,j}_{X\times X,n}(p_1^* z)\big) \\ &= p_{2*}\big(\rho(\alpha) \cup p_1^*\big(\rho^{i,j}_{X,n}(z)\big)\big) \\ &= \rho(\alpha)_*\big(\rho^{i,j}_{X,n}(z)\big). \end{aligned}$$

□

For an effective Chow motive $M = (X, \pi)$, we define $H^a_{\text{ét}}(M, \mu_n^{\otimes b}) := \rho(\pi)_* H^a_{\text{ét}}(X, \mu_n^{\otimes b})$. By Lemma 2.3, we have a étale cycle map for M;

$$\rho^{i,j}_{M,n} : CH^i(M, j)/n \to H^{2i-j}_{\text{ét}}(M, \mu_n^{\otimes i}).$$

Lemma 2.4 ([Yam05, Lemma 2.5]). *For $t \geq 1$, a k-rational point P in C gives*

$$H^s_{\text{ét}}(h^0(C), \mu_n^{\otimes t}) \simeq H^s_{\text{ét}}(k, \mu_n^{\otimes t}),$$
$$H^s_{\text{ét}}(h^1(C), \mu_n^{\otimes t}) \simeq H^{s-1}_{\text{ét}}(k, J[n] \otimes \mu_n^{\otimes (t-1)}),$$
$$H^s_{\text{ét}}(h^2(C), \mu_n^{\otimes t}) \simeq H^{s-2}_{\text{ét}}(k, \mu_n^{\otimes (t-1)}).$$

Proof. Put $\alpha := [P \times C]$, ${}^t\alpha := [C \times P]$, and $\pi := 1_C - \alpha - {}^t\alpha$. Thus

$$h^0(C) = (C, \alpha), \qquad h^1(C) = (C, \pi), \qquad h^2(C) = (C, {}^t\alpha).$$

Since $h^0(C) \simeq (\mathrm{Spec}(k), 1_k), h^2(C) \simeq \mathbb{L}$, the statement is clear for $h^0(C)$ and $h^2(C)$. Indeed, for $\rho(\alpha) = \rho^{1,0}_{C\times C,n}([P \times C])$, we have

$$\rho(\alpha)_* : H^s_{\text{ét}}(C, \mu_n^{\otimes t}) \xrightarrow{i^*} H^s_{\text{ét}}(k, \mu_n^{\otimes t}) \xrightarrow{f^*} H^s_{\text{ét}}(C, \mu_n^{\otimes t})$$

and

$$\rho({}^t\alpha)_* : H^s_{\text{ét}}(C, \mu_n^{\otimes t}) \xrightarrow{f_*} H^{s-2}_{\text{ét}}(k, \mu_n^{\otimes (t-1)}) \xrightarrow{i_*} H^s_{\text{ét}}(C, \mu_n^{\otimes t}),$$

where $i : P \hookrightarrow C$ is the closed immersion and $f : C \to \mathrm{Spec}(k)$ is the structure map.

For $h^1(C)$, we use the Hochschild–Serre spectral sequence

$$E_2^{a,b} = H^a_{\text{ét}}\big(k, H^b_{\text{ét}}(\overline{C}, \mu_n^{\otimes t})\big) \Longrightarrow H^{a+b}_{\text{ét}}\big(C, \mu_n^{\otimes t}\big).$$

See [Mil13, Theorem 12.7 and Example 12.8] and [NSW08, Chapter II, Section 4, Theorem 2.4.1]. This spectral sequence is multiplicative; on its E_2-page the product is induced by the cup products in continuous Galois cohomology and geometric étale cohomology. See [HS53, pp. 118–119] and [NSW08, Chapter I, Section 4].

The Kummer sequence and the trace isomorphism give

$$H^0_{\text{ét}}(\overline{C}, \mu_n^{\otimes t}) \simeq \mu_n^{\otimes t}, \ H^1_{\text{ét}}(\overline{C}, \mu_n^{\otimes t}) \simeq J[n] \otimes \mu_n^{\otimes (t-1)}, \text{ and } H^2_{\text{ét}}(\overline{C}, \mu_n^{\otimes t}) \simeq \mu_n^{\otimes (t-1)};$$

see [Mil13, Proposition 14.2 and Theorem 24.1] and [Fu15, Sections 8.2–8.3].

The actions of α and ${}^t\alpha$ are compatible with the Hochschild–Serre spectral sequence. On the E_2-page, their actions are induced by the actions of the corresponding correspondences on $H^b_{\text{ét}}(\overline{C}, \mu_n^{\otimes t})$. After base change to $\overline{k}$, the same formulas as above show that $\rho(\alpha)_*$ is the identity on $H^0_{\text{ét}}(\overline{C}, \mu_n^{\otimes t})$ and is zero on $H^1_{\text{ét}}(\overline{C}, \mu_n^{\otimes t})$ and $H^2_{\text{ét}}(\overline{C}, \mu_n^{\otimes t})$. Similarly, $\rho({}^t\alpha)_*$ is the identity on $H^2_{\text{ét}}(\overline{C}, \mu_n^{\otimes t})$ and is zero on $H^0_{\text{ét}}(\overline{C}, \mu_n^{\otimes t})$ and $H^1_{\text{ét}}(\overline{C}, \mu_n^{\otimes t})$. Since

$$\rho(\pi)_* = 1 - \rho(\alpha)_* - \rho({}^t\alpha)_*,$$

it follows that $\rho(\pi)_*$ acts as zero on $H^0_{\text{ét}}(\overline{C}, \mu_n^{\otimes t})$ and $H^2_{\text{ét}}(\overline{C}, \mu_n^{\otimes t})$, and as the identity on $H^1_{\text{ét}}(\overline{C}, \mu_n^{\otimes t})$. Therefore the Hochschild–Serre spectral sequence obtained by applying $\rho(\pi)_*$ has only the row $b = 1$, namely

$$E_2^{a,1} = H^a_{\text{ét}}\big(k, J[n] \otimes \mu_n^{\otimes (t-1)}\big).$$

Hence there are no nonzero differentials, and we obtain

$$H^s_{\text{ét}}(h^1(C), \mu_n^{\otimes t}) = \rho(\pi)_* H^s_{\text{ét}}(C, \mu_n^{\otimes t}) \simeq H^{s-1}_{\text{ét}}\big(k, J[n] \otimes \mu_n^{\otimes (t-1)}\big).$$

This shows the assertion for $h^1(C)$. □

For $r \geq 0$, let $\iota^{\text{ét}}_{J,r}$ denote the inclusion

$$\iota^{\text{ét}}_{J,r} \colon H^{r+1}_{\text{ét}}\big(k, J[n] \otimes \mu_n^{\otimes r}\big) \hookrightarrow H^{r+2}_{\text{ét}}\big(C, \mu_n^{\otimes (r+1)}\big) \tag{2.8}$$

induced by the Hochschild–Serre spectral sequence and the Chow–Künneth splitting, with the total-complex convention of [NSW08, Chapter II, Section 2, Definition 2.2.3 and p. 104].

Lemma 2.5. *For every $z \in J(k) = CH_0(C)^0$, one has*

$$\rho^{1,0}_{C,n}(z) = \iota^{\text{ét}}_{J,0}\big(\delta_J(z)\big) \quad \text{in } H^2_{\text{ét}}(C, \mu_n). \tag{2.9}$$

Equivalently, under the identification

$$H^2_{\text{ét}}\big(h^1(C), \mu_n\big) \simeq H^1_{\text{ét}}(k, J[n]),$$

the cycle map

$$\rho^{1,0}_{h^1(C),n} \colon J(k)/n \to H^1_{\text{ét}}(k, J[n])$$

coincides with the Kummer connecting homomorphism δ_J.

Proof. Under the identification $CH^1(C) = \mathrm{Pic}(C)$, the map $\rho^{1,0}_{C,n}$ is the connecting homomorphism associated with the Kummer sequence on C. Since a degree-zero class belongs to the $h^1(C)$-part, the compatibility of the Kummer sequence with the Hochschild–Serre edge morphism gives the commutative diagram

$$\begin{array}{ccc} J(k)/n & \xrightarrow{\rho^{1,0}_{h^1(C),n}} & H^2_{\text{ét}}(h^1(C), \mu_n) \\ \downarrow{\scriptstyle \delta_J} & & \downarrow{\scriptstyle \simeq} \\ H^1_{\text{ét}}(k, J[n]) & = & H^1_{\text{ét}}(k, J[n]). \end{array}$$

This proves the equivalent assertion, and (2.9) follows by including the $h^1(C)$-part into $H^2_{\text{ét}}(C, \mu_n)$. □

Lemma 2.6. *Let $f \colon C \to \mathrm{Spec}(k)$ be the structure morphism. For $\alpha \in H^1_{\text{ét}}(k, J[n])$ and $\beta \in H^r_{\text{ét}}(k, \mu_n^{\otimes r})$, one has*

$$\iota^{\text{ét}}_{J,0}(\alpha) \cup f^*\beta = (-1)^r \iota^{\text{ét}}_{J,r}(\alpha \cup \beta). \tag{2.10}$$

Proof. In the standard total complex for the Hochschild–Serre spectral sequence, the product of classes of bidegrees (a, b) and (a', b') acquires the sign $(-1)^{ba'}$; see [HS53, Chapter II, Section 1] and [NSW08, Chapter II, Section 4, proof of Theorem 2.4.1]. Here α has bidegree $(1, 1)$ and β has bidegree $(r, 0)$. Thus their product acquires the sign $(-1)^{1\cdot r} = (-1)^r$. Under the Chow–Künneth splitting, this is precisely (2.10). □

3. Proof of the main theorem

Proof of Theorem 1.1. By Lemma 2.2, the map $\iota_{J,r}$ is the inclusion of a direct summand. Hence it remains injective after reduction modulo n. From Lemma 2.2 and Lemma 2.4, we therefore have a commutative diagram

$$\begin{array}{ccc} K_r(k; J, \mathbb{G}_m)/n & \xrightarrow[(2.7)]{\iota_{J,r}} & CH^{r+1}(C, r)/n \\ \downarrow{\scriptstyle \rho^{r+1,r}_{h^1(C),n}} & & \downarrow{\scriptstyle \rho^{r+1,r}_{C,n}} \\ H^{r+1}_{\text{ét}}\big(k, J[n] \otimes \mu_n^{\otimes r}\big) & \xrightarrow[(2.8)]{\iota^{\text{ét}}_{J,r}} & H^{r+2}_{\text{ét}}\big(C, \mu_n^{\otimes (r+1)}\big). \end{array}$$

The right vertical map is injective by [GL01, Corollary 1.2] (see also [Voe11, Theorem 6.17(1)]); hence the left vertical map

$$\rho^{r+1,r}_{h^1(C),n} \colon K_r(k; J, \mathbb{G}_m)/n \longrightarrow H^{r+1}_{\text{ét}}\big(k, J[n] \otimes \mu_n^{\otimes r}\big)$$

is also injective.

It remains to compare the étale cycle map $\rho^{r+1,r}_{h^1(C),n}$ with the Galois symbol map $S^{J,r}_n$. We first consider symbols defined over the base field. Let $f\colon C \to \operatorname{Spec}(k)$ be the structure morphism. For $z \in J(k)$ and $a_1, \dots, a_r \in k^\times$, the compatibility of the cycle map with the intersection product and the cup product gives

$$\begin{aligned}
\iota^{\text{ét}}_{J,r} \circ \rho^{r+1,r}_{h^1(C),n}\big(\{z, a_1, \dots, a_r\}_{k/k}\big) &= \rho^{r+1,r}_{C,n}\big(z \cap a_1 \cap \cdots \cap a_r\big) \\
&= \rho^{1,0}_{C,n}(z) \cup \rho^{1,1}_{C,n}(a_1) \cup \cdots \cup \rho^{1,1}_{C,n}(a_r) \\
&\overset{(2.9)}{=} \iota^{\text{ét}}_{J,0}\big(\delta_J(z)\big) \cup f^*\delta(a_1) \cup \cdots \cup f^*\delta(a_r) \\
&\overset{(2.10)}{=} (-1)^r \iota^{\text{ét}}_{J,r}\big(\delta_J(z) \cup \delta(a_1) \cup \cdots \cup \delta(a_r)\big).
\end{aligned}$$

Thus the desired comparison holds for symbols over k.

Now let E/k be a finite extension, and let $p_E\colon C_E \to C$ be the projection. Write $\iota^{\text{ét}}_{J_E,r}$ for the corresponding inclusion over E. The naturality of the Hochschild–Serre spectral sequence and the compatibility of proper push-forward with trace give a commutative diagram

$$\begin{array}{ccc}
H^{r+1}_{\text{ét}}\big(E, J[n] \otimes \mu_n^{\otimes r}\big) & \xrightarrow{\iota^{\text{ét}}_{J_E,r}} & H^{r+2}_{\text{ét}}\big(C_E, \mu_n^{\otimes(r+1)}\big) \\
\Big\downarrow{\scriptstyle \operatorname{Cor}_{E/k}} & & \Big\downarrow{\scriptstyle (p_E)_*} \\
H^{r+1}_{\text{ét}}\big(k, J[n] \otimes \mu_n^{\otimes r}\big) & \xrightarrow{\iota^{\text{ét}}_{J,r}} & H^{r+2}_{\text{ét}}\big(C, \mu_n^{\otimes(r+1)}\big).
\end{array} \tag{3.1}$$

See [SGA4, Exposé IX, Section 5] and [Fu15, Section 8.2]. Applying the comparison over E and using the compatibility of the cycle map with proper push-forward, we obtain, for $z \in J(E)$ and $a_1, \dots, a_r \in E^\times$,

$$\begin{aligned}
&\iota^{\text{ét}}_{J,r} \circ \rho^{r+1,r}_{h^1(C),n}\big(\{z, a_1, \dots, a_r\}_{E/k}\big) \\
&\quad= \rho^{r+1,r}_{C,n}\Big((p_E)_*\big(z \cap a_1 \cap \cdots \cap a_r\big)\Big) \\
&\quad= (p_E)_* \rho^{r+1,r}_{C_E,n}\big(z \cap a_1 \cap \cdots \cap a_r\big) \\
&\quad= (-1)^r (p_E)_* \iota^{\text{ét}}_{J_E,r}\big(\delta_{J,E}(z) \cup \delta_E(a_1) \cup \cdots \cup \delta_E(a_r)\big) \\
&\quad\overset{(3.1)}{=} (-1)^r \iota^{\text{ét}}_{J,r} \operatorname{Cor}_{E/k}\big(\delta_{J,E}(z) \cup \delta_E(a_1) \cup \cdots \cup \delta_E(a_r)\big) \\
&\quad\overset{(2.1)}{=} (-1)^r \iota^{\text{ét}}_{J,r} \circ S^{J,r}_n\big(\{z, a_1, \dots, a_r\}_{E/k}\big).
\end{aligned}$$

Since such symbols generate $K_r(k; J, \mathbb{G}_m)$ and $\iota^{\text{ét}}_{J,r}$ is injective, we obtain

$$\rho^{r+1,r}_{h^1(C),n} = (-1)^r S^{J,r}_n.$$

Since $\rho^{r+1,r}_{h^1(C),n}$ is injective, the Galois symbol map $S^{J,r}_n$ is injective. □

Remark 3.1 ([HS25, Lemma 3.2], [Hir24])**.** We record a stable-range l-divisibility result for Somekawa K-groups attached to several semi-abelian varieties and copies of $\mathbb{G}_m$. Let $G_1, \dots, G_q$ $(q \geq 1)$ be semi-abelian varieties over k, and let $l \neq \operatorname{char}(k)$ be a prime. Assume that $s = \operatorname{cd}_l(k) < \infty$. If $r \geq s$, then $K(k; G_1, \dots, G_q, \underbrace{\mathbb{G}_m, \dots, \mathbb{G}_m}_{r})$ is l-divisible. Consequently, its quotient modulo l^m is zero for every $m \geq 1$, and the corresponding Galois symbol map modulo l^m is injective.

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